

Project Summary

Breast Cancer Classification Using Machine Learning

Introduction

Machine Learning has become one of the most important technologies in the healthcare sector, where it assists medical professionals in making accurate and timely decisions. Classification is a supervised machine learning technique used to predict categorical outcomes based on input data. In this project, a classification model is developed to predict whether a breast tumor is **malignant (cancerous)** or **benign (non-cancerous)** using the Breast Cancer Wisconsin dataset available in the scikit-learn library.

The primary objective of this project is to understand the complete workflow of a machine learning classification problem. The project demonstrates how to preprocess data, train different classification models, evaluate their performance using multiple evaluation metrics, and handle class imbalance to improve prediction quality. It also provides practical exposure to model comparison and performance analysis using industry-standard techniques.

Objectives

The main objectives of this project are:

- To understand the concept of supervised classification.
- To build a Logistic Regression model for binary classification.
- To compare Logistic Regression with a Decision Tree classifier.
- To evaluate model performance using different classification metrics.
- To understand the importance of Confusion Matrix, Precision, Recall, F1-Score, ROC Curve, and AUC.
- To learn techniques for handling imbalanced datasets using class weights.
- To compare different machine learning models and select the most suitable one based on evaluation metrics.

Dataset Description

The project uses the Breast Cancer Wisconsin dataset provided by the scikit-learn library. The dataset contains numerical features extracted from digitized images of breast tissue samples. These features describe characteristics such as radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and fractal dimension of cell nuclei.

The dataset consists of 569 observations with 30 numerical input features and one target variable. The target variable has two classes:

Benign (Non-Cancerous)

Malignant (Cancerous)

Since the dataset is already cleaned and free from missing values, it provides an excellent starting point for learning classification algorithms.

Project Implementation

The project begins by importing the required Python libraries, including NumPy, Pandas, Matplotlib, and several machine learning modules from scikit-learn. These libraries are used for data manipulation, visualization, model building, and performance evaluation.

The dataset is loaded into a Pandas DataFrame, and an initial analysis is performed to understand its structure, feature names, dimensions, and class distribution. Basic exploratory data analysis helps verify that there are no missing values and provides an overview of the dataset.

Next, the dataset is divided into training and testing sets using the `train_test_split()` function. The training data is used to train the machine learning models, while the testing data is reserved for evaluating their performance on unseen samples.

Since Logistic Regression is sensitive to differences in feature scales, `StandardScaler` is used to normalize all numerical features. Standardization improves model convergence and helps achieve better prediction accuracy.

After preprocessing, a Logistic Regression classifier is trained using the scaled training data. Once the model is trained successfully, predictions are generated for the testing dataset.

To evaluate the model's performance, several classification metrics are calculated. A Confusion Matrix is generated to display the number of correctly and incorrectly classified samples. The Classification Report provides Precision, Recall, F1-Score, and Accuracy for each class. These metrics provide a detailed understanding of the model's strengths and weaknesses beyond simple accuracy.

The Receiver Operating Characteristic (ROC) Curve is then plotted to visualize the trade-off between the True Positive Rate and False Positive Rate. The Area Under the Curve (AUC) score is calculated to measure the overall classification performance. A higher AUC score indicates a better-performing classifier.

The project also demonstrates handling imbalanced datasets using the `class_weight="balanced"` parameter in Logistic Regression. Although the Breast Cancer dataset is only moderately imbalanced, this technique illustrates how machine learning

models can assign higher importance to minority-class samples, improving classification performance when datasets are unevenly distributed.

To compare different classification algorithms, a Decision Tree classifier is trained using the same dataset. Its performance is evaluated using the same classification metrics, allowing a direct comparison between Logistic Regression and Decision Tree. Finally, the accuracies of all models are summarized in a comparison table.

Evaluation Metrics Covered

Several important evaluation metrics are studied and implemented during this project:

- Confusion Matrix
- Accuracy
- Precision
- Recall
- F1-Score
- ROC Curve
- Area Under Curve (AUC)

These metrics provide a comprehensive understanding of classifier performance and demonstrate why accuracy alone is not always sufficient for evaluating machine learning models.

Tools and Technologies Used

The following tools and technologies were used throughout the project:

- Python Programming Language
- Jupyter Notebook
- NumPy
- Pandas
- Matplotlib
- Scikit-learn Machine Learning Library

Learning Outcomes

Through this project, I gained practical experience in implementing supervised machine learning algorithms for binary classification problems. I learned how to preprocess datasets, split data into training and testing sets, standardize numerical features, train Logistic Regression and Decision Tree classifiers, and evaluate model performance using multiple statistical metrics.

I also developed a better understanding of confusion matrices, precision, recall, F1-score, ROC curves, and AUC values. Additionally, I learned the importance of handling

imbalanced datasets and comparing multiple models before selecting the most appropriate one.

Conclusion

This project successfully demonstrates the complete machine learning workflow for a binary classification problem using the Breast Cancer Wisconsin dataset. Logistic Regression proved to be an effective classifier due to its simplicity, interpretability, and high prediction accuracy. The Decision Tree classifier provided an alternative approach, allowing meaningful model comparison.

Overall, the project strengthened my understanding of classification algorithms, model evaluation techniques, feature preprocessing, and handling class imbalance. These concepts form a strong foundation for developing more advanced machine learning applications in healthcare, finance, cybersecurity, and many other real-world domains.